EE 172 – UNIT 1

Principle of Electromechanical Energy Conversion

Short Detailed Notes

1.1 INTRODUCTION TO ELECTROMECHANICAL MACHINES

What is an Electromechanical Machine?

- A device that **links electrical and mechanical systems**
- Provides **reversible energy flow** through its magnetic field
- Can operate on **AC or DC** power
- Can function as a **motor, generator, or converter**

The Energy Balance Equation

The fundamental equation governing all electromechanical machines:

Energy Input = Energy Lost + Increase in Stored Energy + Energy Output

Or in differential form:

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dW(elec input) = dW(loss) + dW(field) + dW(mech output)
```

Motor vs Generator

Motor	Generator
Converts electrical energy → mechanical energy	Converts mechanical energy → electrical energy
Input: Electricity	Input: Mechanical power (from prime mover)
Output: Mechanical power	Output: Electricity

Types of Motion

1. **Linear (Translational)** – straight-line motion
2. **Rotary (Radial)** – circular motion

Key Formula for Force

The force exerted by a mechanical member:

$$f_x = dW(\text{mech output}) / dx$$

Where:

- f_x = force in direction of displacement

- dx = displacement
- dW = change in mechanical output energy

1.2 FORCE OF ALIGNMENT

Two Ways Machines Develop Mechanical Force

1. **Alignment Force** – magnetic attraction/alignment
2. **Interaction Force** – force on current-carrying conductors

What is Force of Alignment?

- Acts in a direction that **increases magnetic stored energy**
- Can be seen as: (i) **Force of attraction** (pulling poles together); (ii) **Lateral force** (aligning poles side-by-side)

How Alignment Force Works

When poles are not perfectly aligned, the force tries to:

1. Bring poles together (reduce air gap) → decreases reluctance
2. Align poles laterally → increases flux and stored energy

1.2.1 Attracted-Armature Relay

Basic Transducer

Definition: An electromechanical machine that:

- Operates at **very low power levels**
- Provides **signals** to activate electronic control devices
- Example: Microphone (converts acoustic → electrical)

Attracted-Armature Relay

- A **translational transducer** (linear motion)
- Example of a **singly-excited** machine (one coil)

Principle of Operation

1. Coil energized by voltage → current flows
2. Flux set up in core and air gap
3. Surfaces adjacent to air gap become magnetized
4. Attraction pulls the armature/plate in direction shown
5. **Change in magnetic field energy produces the force**

Mathematical Analysis

Electrical Power Input:

$$p_e = e \times i$$

Where: e = induced voltage, i = current

Induced Voltage:

$$e = d(N\Phi)/dt = d(Li)/dt = L(di/dt) + i(dL/dt)$$

Therefore:

$$p_e = Li(di/dt) + i^2(dL/dt)$$

Field Stored Energy:

$$W_f = ½Li^2$$
$$p_f = dW_f/dt = Li(di/dt) + ½i^2(dL/dt)$$

Mechanical Power:

$$p_m = p_e - p_{loss} - p_f$$
$$p_m = ½i^2(dL/dt)$$

Force Equation (Final Form):

$$f = -½I^2(dL/dx)$$

Alternative Form (in terms of reluctance):

$$f = ½\Phi^2(dS/dx)$$

Where: S = reluctance, Φ = flux. **Negative sign** indicates force opposes increasing air gap.

Force Between Parallel Magnetic Surfaces

Reluctance of air gap:

$$S = l_g / (\mu_0\mu_rA)$$

For air: $\mu_r = 1$

Derivative of reluctance:

$$dS/dx = 1/(\mu_0A)$$

Final Force Equation:

$$f = ½\Phi^2(1/\mu_0A)$$
$$f = B^2A/(2\mu_0)$$

This is the **force per air gap** between parallel magnetized surfaces.

1.2.2 Force of Interaction

Application in Rotary Machines

Basic Principle: Force on a **current-carrying conductor** in a magnetic field. Used in motors, galvanometers, etc.

Rectangular Coil Analysis

Given:

- N turns of coil
- Dimensions: l × b (length × width)
- Plane of coil at angle θ to magnetic field B

Flux through coil:

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Φ = BA sin θ
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Force on conductor:

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F = B I N sin θ
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Torque on coil:

```
T = F × b
T = B A N I sin θ
```

Where: A = l × b (area of coil), θ = angle between plane of coil and field direction

1.3 ROTATING MACHINES

Why Rotating Machines?

- Linear machines have a drawback: **motion is not continuous**
- Rotating machines provide **continuous rotary motion**

Types of Rotating Machines

Generator	Motor
Mechanical input → Electrical output	Electrical input → Mechanical output
Driven by prime mover (turbine, engine)	Drives mechanical devices (conveyors, tools)

Fleming's Right-Hand Rule (Generator Action)

Used to find **direction of induced EMF**

How to use:

1. Thumb → Direction of motion of conductor
2. Forefinger → Direction of magnetic flux (N→S)
3. Middle finger → Direction of induced EMF

Fleming's Left-Hand Rule (Motor Action)

Used to find **direction of force/motion**

How to use:

1. Forefinger → Direction of magnetic field
2. Middle finger → Direction of current

3. Thumb → Direction of force/motion

Force Magnitude:

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F = BIL
```

Where: B = flux density, I = current, L = active length of conductor

Ratings

- **Generators:** Rated in kW or MW (power delivered)
- **Motors:** Rated in hp or kW (power delivered)

1.4 CONSTRUCTIONAL FEATURES

Essential Parts of Rotating Machines

1. **Stator** – Stationary member
2. **Rotor** – Rotating member
3. **Auxiliary Equipment** – Brushes, commutator, bearings, etc.

1.4.1 Stator Unit

Components:

1. Stator frame/yoke
2. Pole-cores
3. Pole-shoes
4. Field/Exciting coils

Functions of Stator Frame/Yoke:

1. Mechanical support for poles
2. Protection for the machine
3. Carries magnetic flux produced by poles

Functions of Pole-cores and Pole-shoes:

1. Spread out flux in the air gap
2. Reduce reluctance (larger cross-sectional area)
3. Support the field coils

Materials:

- Cast iron or cast steel
- Forged steel
- **Steel laminations** (to reduce hysteresis and eddy-current losses)

1.4.2 Rotor Unit

Construction:

- Cylindrical or drum-shaped
- Built of **steel laminations**
- Has **slots** to house rotor windings (armature)

Rotor Windings (Armature):

- Insulated from each other
- Placed in slots lined with tough insulating material

1.4.3 Auxiliary Equipment

1. Commutator (Split Rings):

- Made of **copper segments**
- Purpose: **Collect current** from armature windings
- Converts AC in armature to DC output (or vice versa)

2. Brushgear:

- Collects current from rotating commutator
- Feeds current to commutator
- Components: brushes, brush holders, brush studs, busbars

3. Armature Shaft Bearings:

- **Ball bearings** – frequently used (reliable)
- **Roller bearings** – for heavy duties
- Lubricated with hard oil for quieter operation and reduced wear

4. Other Equipment:

- Slip rings (collector rings) for AC machines
- Starting circuit (for single-phase AC motors)

1.5 IMPORTANT EQUATIONS SUMMARY

Magnetic Circuit Equations

$mmf = Hl = IN$ (Magnetomotive force)
$B = \mu H = \mu_r \mu_0 H$ (Flux density)
$S = l/(\mu_0 \mu_r A)$ (Reluctance)
$\Phi = mmf/S = NI/S$ (Flux)
$L = N\Phi/I = N^2/S$ (Inductance)
$W_{fd} = \frac{1}{2}LI^2$ (Field stored energy)

Force Equations

Alignment Force:

$f = -\frac{1}{2}I^2(dL/dx)$ (General form)
$f = \frac{1}{2}\Phi^2(dS/dx)$ (Reluctance form)
$f = B^2A/(2\mu_0)$ (Per air gap, parallel surfaces)

Interaction Force:

$F = BIL$ (Force on conductor)
$T = BAN\ I \sin\theta$ (Torque on coil)

1.6 COMMON MISTAKES TO AVOID

1. **Don't confuse generator and motor actions** – use Fleming's rules carefully
2. **Remember the negative sign** in force equation – force opposes increasing air gap
3. **For multiple air gaps**, multiply the force equation by number of gaps
4. **Check units** – convert mm to m, mm^2 to m^2
5. **For air gaps**, relative permeability $\mu_r = 1$

1.7 TIPS FOR SOLVING PROBLEMS

1. **Identify number of air gaps** in the magnetic circuit
2. **Set up force balance** (magnetic force = mechanical force)
3. **Solve for flux density** B using: $B = \sqrt{(2\mu_0 F/A)}$
4. **Find H** using: $H = B/(\mu_0\mu_r)$
5. **Find mmf** using: $mmf = H \times$ total length
6. **Find current** using: $I = mmf/N$

End of Notes